

The Temporal Support Behind One Benchmark Timestamp: A Released Event Detector and DSEC’s Ceiling Exposures

Supplementary Material

Anonymous CVPR submission

Paper ID *****

This document holds the material referred to from the main paper: the errors-in-variables treatment of the velocity regressor, the protocol behind the occlusion measurement, the construction and the standard error of the anisotropic decomposition, the data paths of the DSEC measurements, and the prior-art search record. Section numbers below refer to this document; references of the form “main Sec. n ” refer to the paper.

1 The velocity regressor

The regression of main Eq. 1 takes the residual vector against a label velocity that is finite differenced from labels rather than observed. Write the observed velocity as $v = v^* + \eta$. Three error mechanisms enter, and they behave differently. The third of them is the one that decided the measurement.

Classical dilution. Let the label center carry noise of standard deviation σ_c per frame and let the speed be formed by a central difference over $\pm\Delta$. The differenced speed then carries noise of standard deviation

$$\sigma_\eta = \frac{\sigma_c}{\sqrt{2}\Delta}, \quad (1)$$

and τ is attenuated toward zero by $\lambda_{\text{att}} = \text{Var}(v^*)/(\text{Var}(v^*) + \sigma_\eta^2)$. The noise is the label noise of the dataset the fit runs on, so it is measured there: Sec. 3 solves for $\sigma_c = 0.6515$ px on Gen1 itself, and Eq. 1 is evaluated rather than assumed. On the reported rows $\text{Var}(\eta)$ is 0.6 against a $\text{Var}(v)$ of 110.6 in px^2/s^2 , so $\lambda_{\text{att}} = 0.995$ and the correction moves τ by less than 0.1 ms. The synthetic arm of Sec. 4 injects a known lag under a much larger noise and recovers 0.92 of it against a predicted 0.933, so the correction is calibrated as well as small. The attenuation is toward zero, so it cannot manufacture the null that is reported.

Smoothing bias. A central difference returns the mean velocity over 2Δ rather than the instantaneous velocity. Expanding the trajectory about the query instant,

$$\frac{x(t + \Delta) - x(t - \Delta)}{2\Delta} = \dot{x}(t) + \frac{\Delta^2}{6} \ddot{x}(t) + O(\Delta^4), \quad (2)$$

so the regressor carries a systematic term of order $\Delta^2 \ddot{x}/6$. That is a bias in the regressor, not noise in it, and no errors-in-variables correction removes it. It is made small by making Δ small, which Eq. 1 says makes the dilution worse.

The shared-noise term. The first two mechanisms are properties of the regressor alone. The third couples the regressor to the response. A forward difference $(c_{k+1}^* - c_k^*)/\Delta$ carries the label noise δ_k of the current center with weight $-1/\Delta$, and the residual $\hat{c} - c_k^*$ carries the same δ_k with weight -1 , so

$$\text{Cov}(\hat{c} - c_k^*, v_f) = \frac{\text{Var}(\delta)}{\Delta} \quad (3)$$

even at $\tau = 0$, and the induced slope is $\text{Var}(\delta)/\Delta \text{Var}(v)$. A backward difference carries δ_k with the opposite sign. The centered difference omits c_k^* and is free of the term, which is why main Eq. 1 is fitted on it and why the forward and backward fits are reported beside it as a control.

An instrument that does not hold on this data. An instrumented estimate would replace the current speed by a velocity finite differenced from a strictly disjoint earlier label pair, on the argument that disjointness implies independent errors. The label errors of this family are not white, so that argument does not hold. The third-difference kernel $(1, -3, 3, -1)$ forces $\rho(1) = -0.75$ for white noise; the same code returns -0.740 on white noise and -0.3566 on the DSEC-Det labels, and pooled-ratio bias, heteroscedastic segments and heavy tails each returned the white-noise value through the identical pooling code, so the departure is a property of the labels. That measurement does not identify what the correlation structure is, beyond excluding a first-order autoregressive form. No instrumented estimate is reported in the paper.

The construction term. A benchmark whose high-rate labels are constructed between anchors adds a term to $e_{||}$ that has nothing to do with the predictor. The chord of a curved image trajectory runs ahead of the truth throughout the interval and meets it at the anchors, which produces a signed, frame-locked, resetting residual for a predictor with no lag. Main Sec. 4.1 shows that no such term arises inside the DSEC-Det archive used here, and the Gen1 measurements of main Sec. 3 are taken at label times.

2 The influence profile

Main Sec. 3.3 measures the temporal influence profile of the released `rvt-t` checkpoint. The stored input is a stacked histogram of ten bins of 5 ms covering the 50 ms preceding the query time. For each sample the recurrent state is advanced over eight preceding steps so that the network is not read out of a zero state, the reference output is recorded, and then each bin in turn is zeroed and the output recorded again. A bin’s influence is the magnitude of the change its occlusion produces in the emitted detection tensor, relative to its norm, averaged over 480 samples drawn from the released Gen1 validation sequences. The influence-weighted centroid is $\sum_k a_k c_k / \sum_k a_k$ over the ten bin centers c_k . The occluded pass starts from the recurrent state *entering* the step, not from the state the reference pass returned, so the two share one history and differ only in the occluded bin; from the returned state they run one step out of phase and the centroid moves +1.19 ms.

Two properties of that measurement bound it. Occlusion reports the sensitivity of the output to a bin, and sensitivity equals the linear weight a_k of main Sec. 3.1 only for a predictor linear in its bins; for a non-linear predictor the profile is a local rather than a global description. And a single-bin occlusion leaves the remaining bins in place, so the measurement is a partial derivative rather than a decomposition of the output.

Masking the leading bins of the stored input is not a route to the same quantity at a shorter support. For a fixed network of influence weights a_k on bin centers c_k , the conditional expectation after masking the leading j bins is $A_j p_0 + C_j v$ with $A_j = \sum_S a_k$ and $C_j = \sum_S a_k c_k$ over the retained set S . Reading a time off that requires the normalization C_j/A_j , which the network need not perform, and the result equals the retained-mass centroid only for non-negative weights. No masked measurement enters the paper.

3 The label-centre noise of Gen1

The errors-in-variables treatment of Sec. 2 and the calibration of the velocity estimator both need the dispersion of the Gen1 label centre about a smooth trajectory. The value

solved for elsewhere in this work, 0.4974 px, is DSEC-Det’s, and does not transfer to a different annotation effort on a different sensor, so it is solved again on Gen1.

For a track centre series $x_i = f(t_i) + e_i$ on a uniform grid, the k -th difference has $\text{Var}(D_k) = \binom{2k}{k} \sigma^2$ plus a motion term of order k , so $\text{Var}(D_k)/\binom{2k}{k}$ falls toward σ^2 from above. Halving the sampling rate multiplies the motion term by 4^k and leaves σ^2 untouched, giving a second equation in the same two unknowns, $\sigma^2 = (4^k V_{\text{full}} - V_{\text{half}})/(4^k - 1)$. Tracks are rebuilt by the same same-class overlap linking the output-time measurement uses, giving 691 tracks of at least twelve labels on a uniform grid. Variances are trimmed at 5 %, because all four Gen1 box coordinates are stored as integers and a median absolute deviation on lattice-valued data locks onto discrete values.

The two-rate elimination agrees with the full-rate ladder to four decimals from $k = 5$ onward, which is the check that the motion term has died rather than been assumed away. The ladder extrapolates to $\sigma = 0.6515$ px, of which 0.1443 px is the quantization of a 0.5 px centre lattice, leaving 0.6353 px of annotation dispersion.

4 The sign-flip calibration

Equation 3 predicts an artifact whose size is set by the label noise and the observed velocity variance, so it is calibrated before the estimator is used. 2000 synthetic tracks of fourteen labels on the Gen1 grid are generated with quadratic motion; the simulated detector reports the true position at $t - \tau_{\text{true}}$ with its own independent error, and the label centers are corrupted at 0.6515 px.

With $\tau_{\text{true}} = 0$ the forward estimator returns -5.64 ms and the backward one $+5.84$ ms, summing to $+0.20$ ms, against the closed form -5.63 ms; the centered estimator returns $+0.00$ ms. Setting the label noise to zero returns all three to $+0.02$ ms, and the placebo stays at its null throughout, so it does not absorb the artifact and remains usable on real rows. With a lag injected, the centered estimator recovers 0.92 of it, the attenuation Eq. 1 predicts at 0.933 for this noise.

5 The output-time specification sweeps

Do the controls absorb a lag? A detector lagging by τ displaces its prediction by $-\tau v^*$, so adding exactly that to the observed residual builds a dataset with a known extra lag and every real confound left in place. Under the full control set the injected offset comes back at 1.000 of its size at 10, 25, 50 and 100 ms, which also shows v^* is not

in the span of the controls. Injecting instead a purely geometric displacement of 0.05 px per px of box height moves the minimal specification’s τ by +0.63 ms and the reported one’s by less than 0.01 ms, which is what the controls are for.

Does the axis pairing make the answer? The reported specification pairs box height and image row with $\hat{y}$ and box width and image column with $\hat{x}$. Entering all four on both axes gives -3.62 ± 7.41 ms. Dropping each in turn moves τ by under 0.3 ms except for the image column, whose removal returns the placebo to its inflated value. Raising the overlap floor to 0.7 gives $+0.49 \pm 4.67$ ms, the tightest of the sweep. Across every specification carrying geometric controls, including speed floors from 5 to 20 px/s, τ spans -20.2 to $+11.1$ ms, and none of them reaches the influence centroid.

The bootstrap. Resampling sequences with replacement, 4000 times, gives a 95 % interval of $[-17.0, +11.5]$ ms with 3 draws reaching the influence centroid. The fit absorbs the per-sequence constant vectors by within-cell demeaning, which reproduces the full design’s coefficient to 0.007 ms.

6 The per-object ratio

The estimator main Eq. 1 replaces is the per-object ratio $-\langle \hat{c} - c^*, \hat{u} \rangle / \|v^*\|$. Fitted on the same 22 534 rows it has a sample standard deviation of 4170 ms and an interquartile range of 430 ms. Its mean over all rows is $+24.22 \pm 27.73$ ms, which lies on the influence centroid of main Sec. 3.3 and would have been read as confirming the hypothesis that main Sec. 3.5 excludes. Above a 15 px/s floor it gives $+13.34 \pm 6.17$ ms, significant at more than two standard errors, which is a fixed spatial bias divided by a speed: the ratio has no intercept and main Eq. 1 does.

7 The port of the released checkpoint

The released `rvt-t` checkpoint loads into a model built from the repository’s own configuration files with 0 missing and 0 unexpected keys at 4.41 M parameters, and runs forward under a current framework without the training code it was serialized from. Both predictor measurements run on that port.

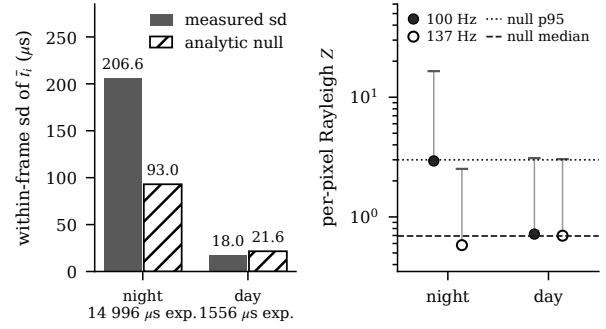


Figure 1: Left: within-frame dispersion of $\bar{t}_i$ against its analytic null in the two exposure strata, 1134 frames of the ceiling sequence at 14 996 μs and 20 day frames at 1556 μs . Right: per-pixel Rayleigh statistic $Z = R^2/N$ at 100.00 Hz and at 137 Hz for the same two sequences, median with a whisker to the ninety-fifth percentile, against the $\text{Exp}(1)$ null. Both contrasts run in the same direction, and Main Sec. 4.4 reports the controls applied to that dispersion.

Arm	frames	sd	null	excess
Ceiling sequence, all pixels	1134	206.6	93.0	183.6
Phase-locked pixels removed	1134	205.1	97.1	180.6
Random pixels, matched size	1133	206.3	94.8	182.9
Random-position boxes	1123	338.2	134.5	306.3
<i>by each box's own 100.00 Hz modulation depth m</i>				
$m \in [0.122, 0.410)$	270	119.3	74.6	93.1
$m \in [0.410, 0.467)$	243	149.1	76.2	128.1
$m \in [0.467, 0.534)$	207	156.6	85.1	131.5
$m \in [0.534, 1.050]$	265	227.2	91.5	208.0

Table 1: Per-object event centroid inside one published exposure window of `zurich.city_09_a`, in μs , each entry the median over frames of a within-frame statistic against that frame’s own null $w^2/12N_i$. The first block’s excess follows main Eq. 3 summarized over frames. Rows two and three remove the 14 334 pixels phase-locked at 100.00 Hz and a random subset of that size; row four places boxes of the same shapes at random positions. In the lower block a frame enters a stratum when 3 of its boxes fall in it, so those frame counts do not partition the first row’s.

8 The DSEC measurements

Data path. For each labeled frame of a sequence, the frame’s own exposure interval $[a, b]$ is read from the dataset’s published `image_exposure_timestamps_left` file, the event stream is sliced to that interval through the release’s `ms_to_idx` index, and the events inside each released box are collected. The `ms_to_idx` index resolves to whole milliseconds, so the slice it returns is filtered to $a \leq t < b$ in microseconds before any statistic is formed, and the analytic null is computed for the same interval. Events and frames share one clock in this dataset, so no calibration step enters. For the dispersion of main Sec. 4.4 a box contributes only if it carries at least 200 events in the window, and a frame only if at least 3 boxes qualify; the reported statistics are medians over frames of a within-frame quantity, so no frame is weighted by its

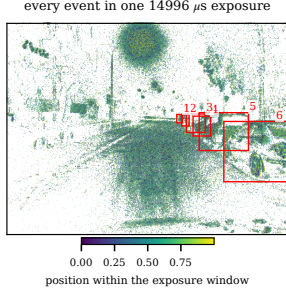


Figure 2: Every event inside one published 14996 μs exposure window of `zurich_city_09_a`, colored by its position within that window, with that frame’s released boxes numbered left to right. The frame is the median-dispersion one of 602 candidates, not the largest.

Sequence	boxes	m	$Z_{100.00}$	Z_{137}	locked
<code>zurich_city_00_a</code>	2914	0.428	117.6	3.15	100.00 %
<code>zurich_city_01_a</code>	2134	0.440	99.1	4.69	99.63 %
<code>zurich_city_02_a</code>	619	0.445	68.4	2.75	99.68 %
<code>zurich_city_03_a</code>	241	0.489	310.9	25.85	95.44 %
<code>zurich_city_09_a</code>	6997	0.466	251.5	7.82	99.23 %
<code>zurich_city_10_a</code>	3611	0.438	111.6	3.33	100.00 %

Table 2: Every DSEC training sequence whose exposure is pinned at 14996 μs , measured per labeled box over the events inside that box during that frame’s own published exposure window, with at least 400 events per box. m is the median over boxes of the modulation depth $m = 2|C|$ at 100.00 Hz, against a floor of 0.089 for 400 events drawn at random phase. $Z_{100.00}$ and Z_{137} are the median over boxes of the Rayleigh statistic at 100.00 Hz and at the off-frequency control on the same events. The analytic $\text{Exp}(1)$ median of 0.693 does not describe either column, which is why the two are compared with each other; *locked* is the fraction of boxes whose local excess at 100.00 Hz exceeds the ninety-ninth percentile of the same quantity at 137 Hz.

object count. For the per-box statistics of Table 2 a box contributes with at least 400 events and no frame-level requirement applies.

The stereo exposures. The left and right exposures open at the same microsecond in 100.00 % of the 7080 frames checked and differ only in when they close, and the published image timestamp equals the mean of the two mid-exposures to within 1 μs .

Which archive, and how tracks are associated. DSEC-Det ships more than one annotation version. The numbers in main Sec. 4.1 describe the archive `dsec-det_left_object_detections.zip`, 4655023 bytes, retrieved 2026-09-01. Whether those labels are tracker output on the RGB frames, which the dataset’s own description states, is not distinguishable from human annotation by a second-difference residual. On the Gen1 side the `track_id` field is a per-frame index rather than a persistent identifier, so association for the output-time fit is rebuilt by same-class overlap between

adjacent label times, at which 79.0 % of boxes link.

Null. For events uniform on $[a, b]$ the mean of N_i times has variance $w^2/12N_i$ with $w = b - a$. Each frame therefore carries its own null, formed from the published width and the observed counts, and the excess of main Eq. 3 is summarized over frames rather than formed from the medians of the two summaries. That is why the excess column of Table 1 is not $(\text{sd}^2 - \text{null}^2)^{1/2}$ of the two columns beside it.

Phase test. The per-pixel test at frequency f takes the event phases $\phi_k = 2\pi f t_k \bmod 2\pi$ of a pixel, forms the resultant length R and reports $Z = R^2/N$, which is $\text{Exp}(1)$ under uniform phase. Raw modulation depth is not usable at these counts, because for N events at random phase the resultant is already of the size of the effect. Pixels enter with at least 100 events over the tested slice, and every fourth event is used, which preserves phase statistics and reduces power, so reported locked fractions are conservative. Two arms hold the estimator to its null, an off-frequency arm on the same events and the same frequency on daytime events. The fraction of locked pixels is quoted of live pixels in main Sec. 4.2 and separately of the sensor; the two denominators differ and each statement names its own.

The per-box statistics of Table 2. The same estimator is applied to the events inside one released box during one published exposure window rather than to the events of one pixel over a long slice. For box i the depth is $m = 2|C|$ and the statistic is $Z = N_i|C|^2$ with $C = N_i^{-1} \sum_{k \in i} e^{-i\omega t_k}$, phases referred to that frame’s mid-exposure instant. The null is unchanged, since Z is $\text{Exp}(1)$ under uniform phase whatever the count. The floor 0.089 quoted against the depths of Table 2 is the expected depth of 400 events at random phase; main Sec. 4.4’s 0.125 is the same expression at its 200-event floor. Boxes are matched to exposure windows by nearest mid-exposure instant within half a frame period. The off-frequency arm at 137 Hz is run on exactly the same events.

Pixel exclusion and modulation strata. The exclusion mask of Table 1 is the set of pixels reaching $p < 10^{-3}$ in the per-pixel test, and the matched control draws a random pixel set of exactly the same size, so the two arms remove the same number of pixels. The mask is built from a six-second slice at every fourth event. The stratification uses no mask: the quartiles of m are taken over all 6841 qualifying boxes, and a frame contributes to a stratum when at least 3 of its boxes fall inside it, which is why the frame counts of that block are not a partition of the first row’s.

9 The permutation null

The analytic null $w^2/12N_i$ of main Eq. 3 is the variance of a mean of N_i draws uniform on the exposure, so it assumes the events of a box arrive independently and uniformly. They arrive in bursts, and for a different statistic that burstiness inflates the reference distribution by a factor of 55 (main Sec. 4.3).

The null used instead assumes nothing about arrival statistics. For each frame the events of all qualifying boxes are pooled and reassigned to boxes at random, preserving each box’s count. The exposure, the flicker and the scene’s burstiness all survive that reassignment; only which box an event belongs to is destroyed. Over 1134 frames and 200 draws each it returns $78.6 \mu\text{s}$, 0.85 times the analytic null rather than above it, and 94.1 % of frames exceed their own. A burst common to the frame displaces every $\bar{t}_i$ in the same direction and cancels in their dispersion, which is why the statistic was already insensitive to it. Boxes of the same shapes at random positions give $339.4 \mu\text{s}$ against a permutation null of $113.3 \mu\text{s}$, the same conclusion the analytic null gives.

10 Prior-art search record

The searches behind the coverage statement in the main paper were: the citation graph of the longitudinal-decomposition paper, queried through a citation graph API within one call, which returned 40 citing papers; and keyword sweeps of a preprint server over streaming perception, temporal tolerance, effective latency, and aggregation windows. Of the returned citing papers, 6 mention time, latency, streaming or event sensing in their title or the opening of their abstract, and 0 apply the decomposition to a temporal axis or to an event camera. The one returned title whose framing could have concerned latency rather than multi-frame feature aggregation has since been retrieved and read, and the main paper states what it does: it scores compute latency on named hardware and evaluates on nuPlan.

Two limits are carried wherever that count is cited. A follow-up request for the total citation count was rate limited, so the returned set is a sample of unknown coverage rather than a census, and keyword filtering on a title and the opening of an abstract will miss a paper that does the relevant thing without saying so early. We did not sweep the engineering-society digital libraries, where the closest ancestor of our estimator was published; the claim is that we searched and did not find, in the places named, not that the work does not exist.

11 The carried state of the SSM release

Main Sec. 3.3 measures five released checkpoints of two architectures. The second is SSM-ViT (Zubic et al., CVPR 2024), a fork of the RVT repository in which each stage’s ConvLSTM is replaced by an S5 layer; it distributes RVT’s own preprocessed Gen1 and builds its window boundaries with the same statement of the same script, so the window convention of main Sec. 3.2 holds there unchanged.

Its recurrent state does not. That release injects the state carried into a chunk as $\bar{\Lambda}_0 \leftarrow \bar{\Lambda}_0 \odot s$ before an associative scan whose binary operator is $(a_i, b_i) \circ (a_j, b_j) = (a_j a_i, a_j b_i + b_j)$. The scan’s outputs are the b components, and b_p is built from $a_1, \dots, a_p$ and $b_0, \dots, b_p$; $\bar{\Lambda}_0$ enters only the a component of the first element and therefore no output. The carried state is inert.

That is a statement about code, so it is measured. Driving each model the way its own release drives it and zeroing the state entering a chunk of 8 windows, the relative L_2 change in the emitted detection tensor is 0.00000 at every position for SSM-ViT, and 0.0616 at position one falling to 0.0304 at the last position for `rvt-t`. A model that ignored its history altogether would return zero to both arms, so the control occludes the window one position earlier inside the same chunk: SSM-ViT answers 0.0276 there, falling to 0.0106, so it is recurrent within a chunk and only across chunks is the state lost.

12 Does the interval change a comparison?

Main Sec. 3.6 reports that displacing the ground truth to the newest window’s centroid costs 0.04 points on one checkpoint. Whether the interval changes a *comparison* needs more than one detector, so all 5 released checkpoints of main Sec. 3.3 were run over the Gen1 validation split under one protocol: the same frames, the same confidence threshold and non-maximum suppression, and ground truth that is identical across the five and identical to the dump main Sec. 3.6 already uses. Each was then scored at the label instant and again at the displacement it prefers, which ranges from -15 to 0 ms across the five.

The ordering cannot change under alignment, and that is arithmetic rather than a failure to find an effect: in every stratum the largest gain any model draws from displacement is 52 % of the smallest gap between adjacent models, at most 0.14 points against 0.11. No refinement of the displacement grid reorders them. The aggregate number and the fast-object number do name different winners, which is a reason to report the strata, but the cause is object speed and not the temporal support this paper measures.

stratum	ordering at the label instant
all moving	$b > s > t > S5-B > S5-S$
10–25 px/s	$S5-B > b > t > S5-S > s$
25–50 px/s	$S5-B > b > S5-S > s > t$
> 50 px/s	$b > t > S5-S > s > S5-B$

Table 3: Ordering of the 5 released checkpoints at the label instant, by object speed; t , s , b are the RVT capacities and $S5-S$, $S5-B$ the SSM ones. Scoring each at its own preferred displacement instead leaves every one of these orderings unchanged and reverses no pair. Speed does what alignment does not: `s5vit-base` is first in the middle two strata and last in the fastest, and `rvt-t` is third overall, last between 25 and 50 px/s and second above 50.

The withdrawn spatial statistic. Main Sec. 6 records that the one spatial statistic chosen for its immunity to a spatially uniform source does not survive its own background control. The two measurements behind that sentence are here. An equal-area annulus of each box’s surroundings aligns with the direction of travel at least as well as the object itself in four of the boxes, and eroding each box to its object-dominated core drives the paired object-minus-annulus difference to -0.178 at 6.2 standard errors, the wrong sign for an object-borne effect. No spatial per-object timing statistic is reported in the paper.

The released streaming evaluation steps through the split in non-overlapping chunks of 21 windows (`sequence_length` in `config/dataset/gen1.yaml`; `start_indices` in `sequence_for_streaming.py` steps by exactly that). The temporal support of a released detection therefore depends on where its frame falls inside that chunk, from a full 21 windows at the last position to the current window alone at the first. This is reported as a property of the release that its documentation does not state, which is the same class of undeclared quantity the rest of this paper measures, and not as a claim about what the architecture could support if the state were carried.